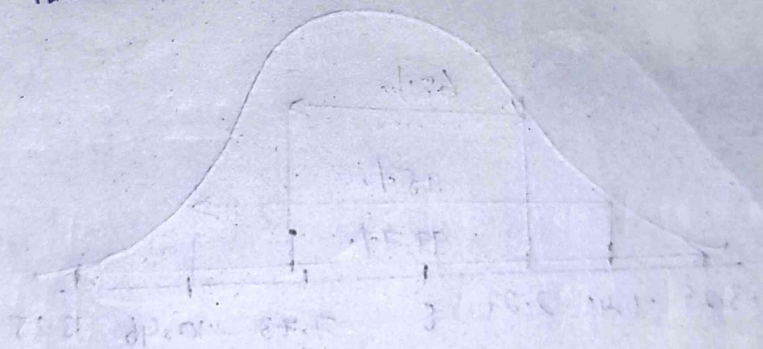


(10) Probability

→ Statistics and Probability go hand in hand, Cause Statistics is all about data & analysing data.

→ Probability is going to allow us to determine how reliable our statistical results actually are.

- ① Basic probability concepts
- ② Types of events
- ③ probability rules.
- ④ Conditional probability
- ⑤ Bayes Theorem



Basic

A probability will always comes b/w 0 to 1. and tells the likelihood of an event occurring. can also tell in it percentage form or form of fraction.

$$0.26 \times 100 = 26\%$$

Calculate the Probability:-

$$P(H) = \frac{\text{True outcome}}{\text{Total outcome}}$$

Ex:- Coin flip

$$P(H) = \text{Head}, P(T) = \text{Tail}$$

$$\Rightarrow P(H) = \frac{1}{2}, P(T) = \frac{1}{2}$$

→ What is the probability of getting heart in deck of cards:-

13 hearts, 13 clubs, 13 diamond, 13 spades

$$52 \text{ Cards}$$

$$\Rightarrow \frac{13}{52} = \frac{1}{4}$$

1) Independent events :- In probability, Independent events are events where the occurrence of one event does not effect the occurrence of other.

→ Mathematically, two events A & B are independent,

$$P(A \cap B) = P(A) \cdot P(B)$$

→ This means the probability of both events happening together is simply the product of their individual probabilities.

Ex :- Tossing a Coin :-

→ Let A be the event that the first toss is heads.

→ Let B be the event that the second toss is Tails.

→ These two events are independent because the result of the first toss does not affect the second.

$$\Rightarrow [P(A \cap B) = P(A) \cdot P(B)] \quad P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2}$$

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

→ Dependent Events :- In probability Dependent events are events where the occurrence of one event is affected the probability of the other.

Mathematically, two events A & B are dependent if

$$[P(A \cap B) \neq P(A) \cdot P(B)]$$

(or)

$$[P(B|A) \neq P(B)]$$

Ex: Drawing cards without Replacement:-

Suppose you have a deck of 52 cards

→ Let A be the event of drawing an Ace on the first draw.

→ Let B be the event of drawing another Ace, on the second draw without putting first card back.

→ Since the card is not replaced, the total number of cards reduces to 51.

→ So Probability of B is affected by A. They are dependent.

$$P(A \cap B) = P(A) \times P(B|A)$$

$$= \frac{4}{52} \times \frac{3}{51}$$

→ Mutually exclusive events are events that cannot happen at the same time.

→ Two events A & B are mutually exclusive if -

$$P(A \cap B) = 0.$$

→ That means the probability that Both A & B happen together is 0.

Ex:- Tossing a Coin

- Event A → getting heads
- Event B → getting Tails.

You can't get both at the same toss, so A & B are mutually exclusive.

→ Rolling A Die

- Event A → Getting a 3
- Event B → getting a 5

A single die can show only one event - both can't happen together. So these events are mutually exclusive.